Version: 1.0



Selection

C-Preprocessor

Summary

Explore the C preprocessor by defining headers, macros, and conditional directives to enhance code modularity and readability.

#C #Preprocessor #Macros



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Chapter 1

Instructions

- Only this page will serve as a reference: do not trust rumors.
- Watch out! This document could potentially change up until submission.
- Make sure you have the appropriate permissions on your files and directories.
- You have to follow the submission procedures for all your exercises.
- Your exercises will be checked and graded by your fellow classmates.
- Additionally, your exercises will be checked and graded by a program called Moulinette.
- Moulinette is very meticulous and strict in its evaluation of your work. It is entirely automated, and there is no way to negotiate with it. So, to avoid bad surprises, be as thorough as possible.
- Moulinette is not very open-minded. It won't try to understand your code if it doesn't adhere to the Norm. Moulinette relies on a program called `norminette` to check if your files respect the norm. TL;DR: it would be foolish to submit work that doesn't pass `norminette`'s check.
- These exercises are carefully laid out by order of difficulty - from easiest to hardest. We `will not` consider a successfully completed harder exercise if an easier one is not perfectly functional.
- Using a forbidden function is considered cheating. Cheaters get -42, and this grade is non-negotiable.
- You'll only have to submit a main() function if we ask for a program.
- Moulinette compiles with these flags: -Wall -Wextra -Werror, and uses `cc`.
- If your program doesn't compile, you'll get `0`.
- You cannot leave any additional files in your directory other than those specified in the subject.
- Got a question? Ask your peer on the right. Otherwise, try your peer on the left.
- Your reference guide is called `Google / man / the Internet / ...`.
- Check out the Slack Piscine.

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- Examine the examples thoroughly. They could very well call for details that are not explicitly mentioned in the subject...
- By Odin, by Thor! Use your brain!!!



Do not forget to add the *standard 42 header* in each of your .c/.h files. Norminette checks its existence anyway!



Norminette must be launched with the *-R CheckForbiddenSourceHeader* flag. Moulinette will use it too.

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Context

The C Piscine is intense. It's your first big challenge at 42 — a deep dive into problem-solving, autonomy, and community.

During this phase, your main objective is to build your foundation — through struggle, repetition, and especially **peer-learning** exchange.

In the AI era, shortcuts are easy to find. However, it's important to consider whether your AI usage is truly helping you grow — or simply getting in the way of developing real skills.

The Piscine is also a human experience — and for now, nothing can replace that. Not even AI.

For a more complete overview of our stance on AI — as a learning tool, as part of the ICT curriculum, and as a growing expectation in the job market — please refer to the dedicated FAQ available on the intranet.

Main message

- 👉 Build strong foundations without shortcuts.
- 👉 Really develop tech & power skills.
- 👉 Experience real peer-learning, start learning how to learn and solve new problems.
- 👉 The learning journey is more important than the result.
- 👉 Learn about the risks associated with AI, and develop effective control practices and countermeasures to avoid common pitfalls.

Learner rules:

- You should apply reasoning to your assigned tasks, especially before turning to AI.
- You should not ask for direct answers to the AI.
- You should learn about 42 global approach on AI.

Phase outcomes:

Within this foundational phase, you will get the following outcomes:

- Get proper tech and coding foundations.
- Know why and how AI can be dangerous during this phase.

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Comments and example:

- Yes, we know AI exists — and yes, it can solve your activities. But you're here to learn, not to prove that AI has learned. Don't waste your time (or ours) just to demonstrate that AI can solve the given problem.
- Learning at 42 isn't about knowing the answer — it's about developing the ability to find one. AI gives you the answer directly, but that prevents you from building your own reasoning. And reasoning takes time, effort, and involves failure. The path to success is not supposed to be easy.
- Keep in mind that during exams, AI is not available — no internet, no smartphones, etc. You'll quickly realise if you've relied too heavily on AI in your learning process.
- Peer learning exposes you to different ideas and approaches, improving your interpersonal skills and your ability to think divergently. That's far more valuable than just chatting with a bot. So don't be shy — talk, ask questions, and learn together!
- Yes, AI will be part of the curriculum — both as a learning tool and as a topic in itself. You'll even have the chance to build your own AI software. In order to learn more about our crescendo approach you'll go through in the documentation available on the intranet.

✓ Good practice:

> I'm stuck on a new concept. I ask someone nearby how they approached it. We talk for 10 minutes — and suddenly it clicks. I get it.

✗ Bad practice:

> I secretly use AI, copy some code that looks right. During peer evaluation, I can't explain anything. I fail. During the exam — no AI — I'm stuck again. I fail.

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Chapter 2

Foreword

Interesting Random Facts

- **Honey Bees:** Honey bees have hair on their eyes. It helps them collect pollen.
- **Octopuses:** Octopuses have three hearts. Two pump blood to the gills, while the third pumps it to the rest of the body.
- **Space:** A day on Venus is longer than a year on Venus. It takes about 243 Earth days to rotate once on its axis.
- **Bananas:** Bananas are slightly radioactive. They contain a small amount of potassium-40, a radioactive isotope.
- **Human Body:** The shortest war in history lasted only 38 minutes between Britain and Zanzibar in 1896.

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Chapter 3

Exercise 0: ft.h

	Exercise: 0	
ft.h		
Directory: *ex0/*		
Files to Submit: `ft.h`		
Authorized: `None`		

- Create your `ft.h` file.
- It should countain the prototypes of all the following functions:

ft.h

```
void    ft_putchar(char c);
void    ft_swap(int *a, int *b);
void    ft_putstr(char *str);
int     ft_strlen(char *str);
int     ft_strcmp(char *s1, char *s2);
```

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Chapter 4

Exercise 1: ft_boolean.h

	Exercise: 1	
	ft_boolean.h	
Directory: *ex1/*		
Files to Submit: `ft_boolean.h`		
Authorized: `None`		

Create a `ft_boolean.h` file. It'll compile and run the following main appropriately:

main.c

```
#include "ft_boolean.h"

void ft_putstr(char *str)
{
    while (*str)
        write(1, str++, 1);
}

t_bool ft_is_even(int nbr)
{
    return ((EVEN(nbr)) ?  TRUE : FALSE);
}

int main(int argc, char **argv)
{
    (void)argv;
    if (ft_is_even(argc - 1) == TRUE)
        ft_putstr(EVEN_MSG);
    else
        ft_putstr(ODD_MSG);
    return (SUCCESS);
}
```

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Check the norm..

The program should display:

```
?> ./a.out
I have an even number of arguments.
?> # or
?> ./a.out
I have an odd number of arguments.
?>
```



Norminette must be launched with the *-RCheckDefine* flag. Moulinette will use it too.

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Chapter 5

Exercise 2: ft_abs.h

	Exercise: 2	
ft_abs.h		
Directory: *ex2/*		
Files to Submit: `ft_abs.h`		
Authorized: `None`		

- Create a macro `ABS` which replaces its argument by it absolute value:

Macro:

```
#define ABS(Value)
```



Norminette must be launched with the *-RCheckDefine* flag. Moulinette will use it too.

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Chapter 6

Submission and peer-evaluation

Turn in your assignment in your `Git` repository as usual. Only the work inside your repository will be evaluated during the defense. Don't hesitate to double check the names of your files to ensure they are correct.



You need to return only the files requested by the subject of this project.

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